Lab 5: Dynamic Parallellism and Asynchronous Operation
Geavanceerde Computerarchitectuur

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1 Introduction

This lab was on synchronous and asynchronous operations. With synchronous operations the CPU halts its operations and waits until the GPU has finished before going on with its own operations. This means the CPU and GPU are synchronized and the CPU remains idle during the GPU's processing. While this ensures data integrity and consistency, it may lead to less than optimal resource utilization as the CPU waits for the GPU. With asynchronous operations the CPU doesn't wait until the GPU is finished but continues with its own processing. This asynchronous behavior enables better utilization of system resources, as the CPU and GPU can operate in parallel.

Dynamic parallelism was also introduced, but there was no task on this as it is not compatible with the GPU's of the PC's in the lab. Dynamic parallelism refers to the capability of a GPU to launch additional parallel work from within a running kernel, it allows the GPU to dynamically create and launch new parallel workloads without intervention from the CPU. This can be beneficial for applications with complex control flow, irregular data structures, or algorithms that require recursive or iterative computations.

2 Results

2.1 Synchronous operations

For the synchronous operations we implemented the sum, max and multiply kernels with reduction. Each operation was made with its own independent array that is generated the moment before the GPU is called. This way there is a CPU operation followed by a GPU operation.

2.2 Asynchronous operations

For the asynchronous operations we adjusted the synchronous code so that the GPU could operate in parallel to the CPU.

As can be seen in figure 1 there is a minimal difference between synchronous and asynchronous operations. Because of this we timed every CPU operation and every GPU operation, this can be seen in figure 2. The GPU time keeps steady while the CPU time rises with the increasing array size, because it cant compute in parallel and creating a longer array takes more time.

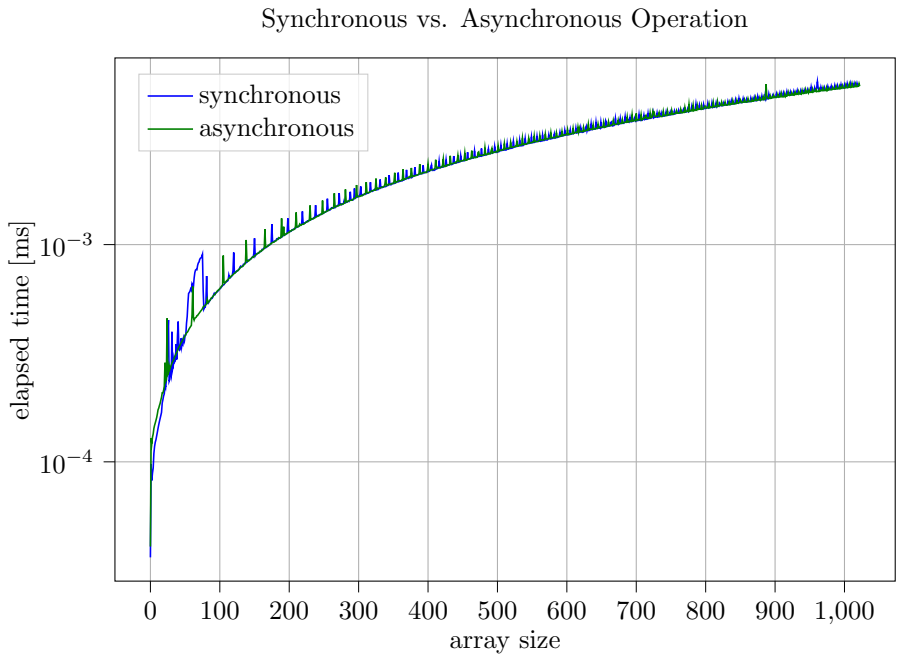


Figure 1: Synchronous vs. Asynchronous Operation

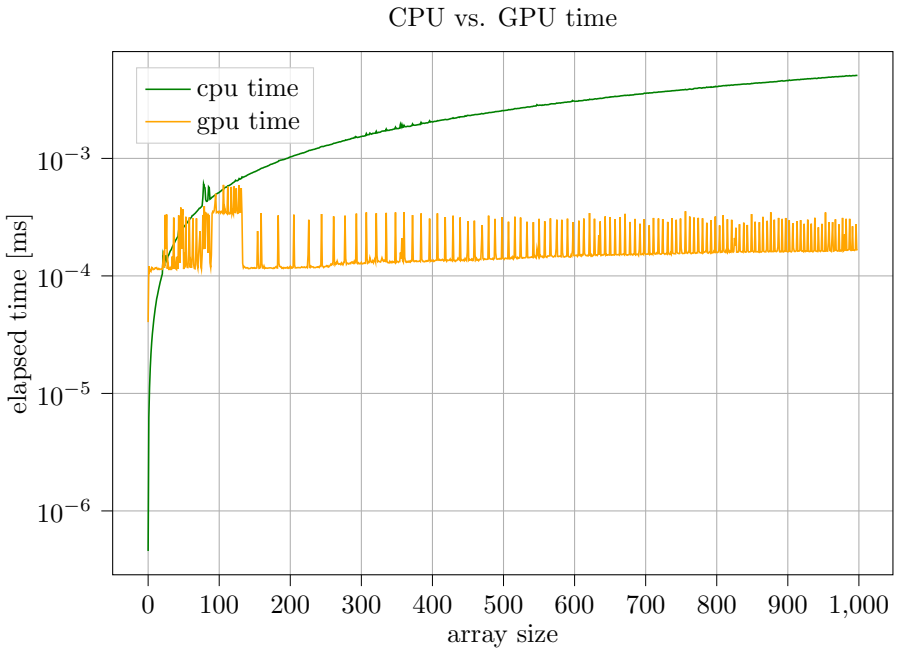


Figure 2: CPU vs.GPU time

A Program

The code can be found at `https://github.com/SienPeerlinck/Geavanceerde-Computerarchitectuur/blob/b1b58965d2dc7e81d3780ecbf84c2cafedae46ef/lab5_ThSi.cu`